STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

15 October 2026

Self-report

To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).

Part 1

1a)

Since $\mathbf{m}$ and $\boldsymbol{\varepsilon}$ are independent and Gaussian, $(\mathbf{m}, \mathbf{d})$ is a linear transformation of a Gaussian vector and therefore jointly Gaussian, with

$$\begin{aligned} E(\mathbf{d}) &= \mathbf{A}\boldsymbol{\mu}_m, \\ \mathrm{Cov}(\mathbf{d}) &= \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I} \;=:\; \boldsymbol{\Sigma}_d, \\ \mathrm{Cov}(\mathbf{d}, \mathbf{m}) &= \mathrm{Cov}(\mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\boldsymbol{\Sigma}_m, \end{aligned}$$

so

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_m \\ \mathbf{A}\boldsymbol{\mu}_m \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_m\mathbf{A}^\top \\ \mathbf{A}\boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_d \end{pmatrix} \right).$$

The conditional distribution of a partitioned Gaussian gives the posterior $\mathbf{m} \mid \mathbf{d} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ with

$$\begin{aligned} \boldsymbol{\mu}_{m|d} &= \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m), \\ \boldsymbol{\Sigma}_{m|d} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}\mathbf{A}\boldsymbol{\Sigma}_m. \end{aligned}$$

The pointwise 95% credibility interval for m_i is $(\boldsymbol{\mu}_{m|d})_i \pm 1.96\sqrt{(\boldsymbol{\Sigma}_{m|d})_{ii}}$.

1b)

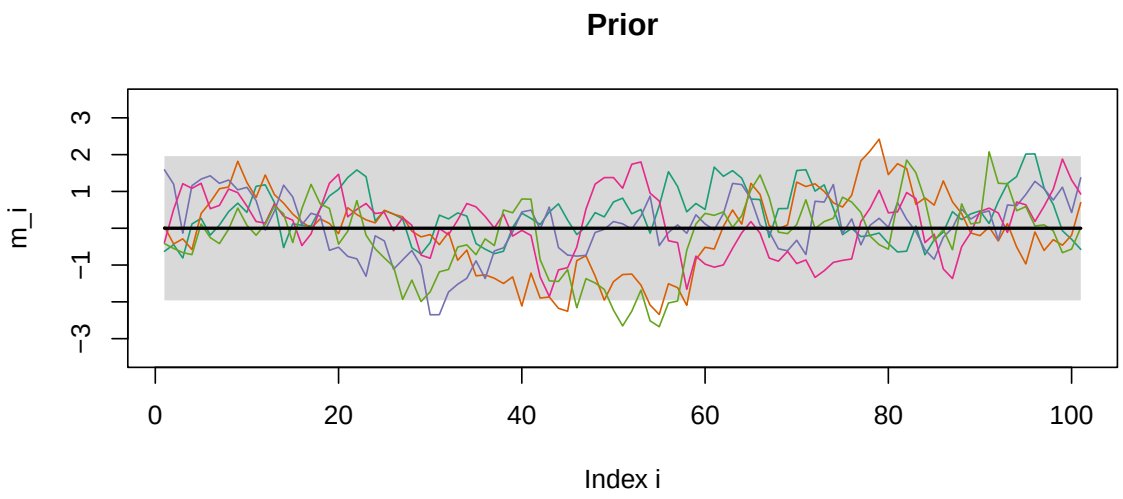


Figure 1: Prior expectation (black), 95% credibility interval (grey) and five simulated profiles.

Since $\Sigma_{m,ii} = 1$, the prior is $m_i \sim N(0, 1)$ and the interval is 0 ± 1.96 for all i. The profiles are simulated as $\boldsymbol{\mu}_m + \mathbf{L}\boldsymbol{x}$, with $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}_m$ (Cholesky) and $\boldsymbol{x} \sim N(\mathbf{0}, \mathbf{I})$.

1c)

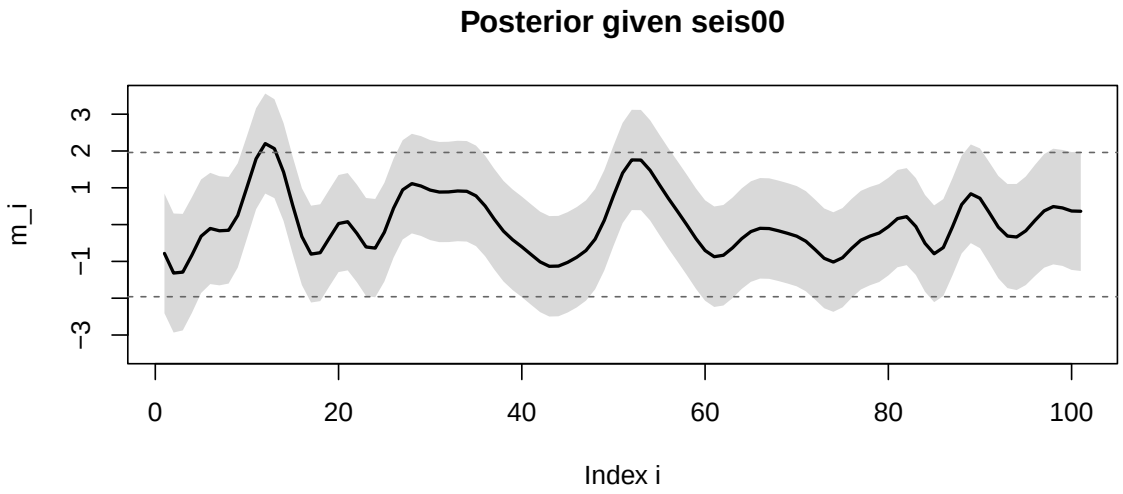


Figure 2: Posterior expectation (black) and 95% credibility interval (grey) given seis00. Dashed: prior interval.

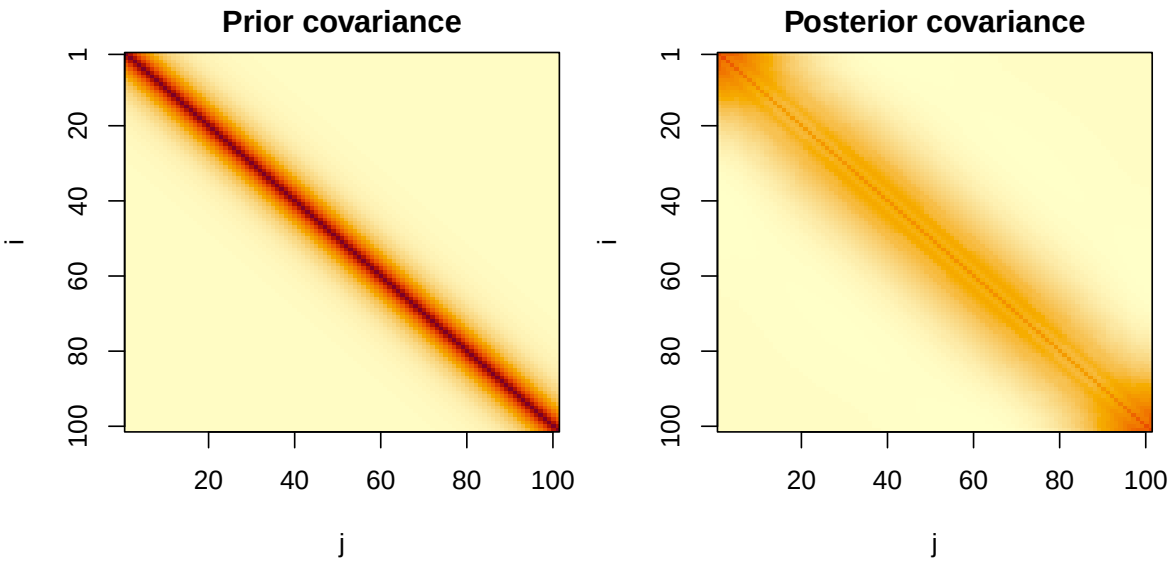


Figure 3: Prior (left) and posterior (right) covariance matrices, same colour scale.

The posterior standard deviations are 0.67–0.83, compared with 1 in the prior. The uncertainty is reduced, but much remains, and the posterior covariance is spread further from the diagonal. One reason is that $\mathbf{A1} = \mathbf{0}$ (the rows of $\mathbf{A}$ sum to zero). The data therefore contain no information about the overall level of $\mathbf{m}$, only about local changes.

1d)

Table 1: Posterior summaries of V ($B = 100$).

mean	median	sd
8.76	5.48	8.68

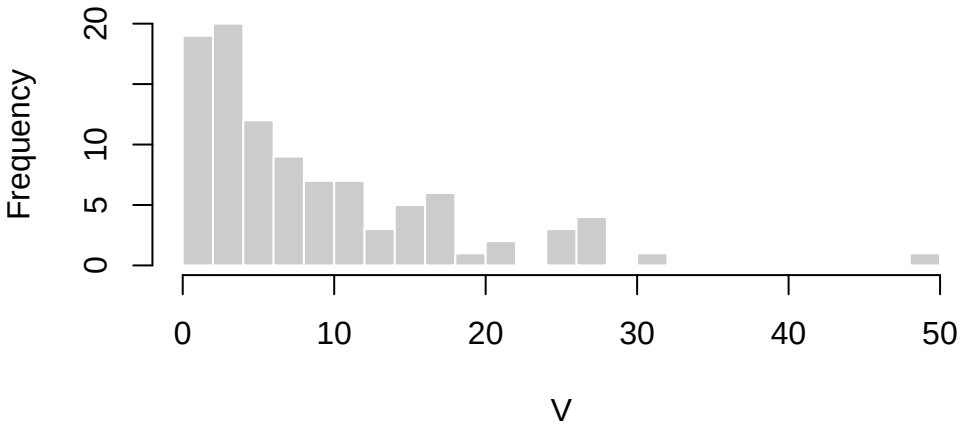


Figure 4: Histogram of 100 posterior samples of V.

We draw $\mathbf{m}^{(b)} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$, $b = 1, \ldots, 100$, and compute $V^{(b)} = h(\mathbf{m}^{(b)})$. The posterior of V is right-skewed, so the mean is larger than the median.

1e)

Estimates from the ordered samples: $\widehat{P}_{10} = V_{(10)} = 0.92$ and $\widehat{P}_{90} = V_{(90)} = 20.64$.

The number of samples below P_{10} is $N \sim \text{Bin}(100, 0.1)$. Since $V_{(r)} \leq P_{10} < V_{(s)}$ exactly when $r \leq N \leq s - 1$,

$$P\big(V_{(r)} \leq P_{10} < V_{(s)}\big) = \sum_{k=r}^{s-1} \binom{100}{k} 0.1^k 0.9^{100-k}.$$

With $r = 5$ and $s = 17$ this probability is 0.956, which gives the interval $[V_{(5)}, V_{(17)}] = [0.19,\ 1.82]$ for P_{10}. The interval is wide compared with the estimate, so $B = 100$ is not sufficient for a precise P_{10}, and B should be increased.

1f)

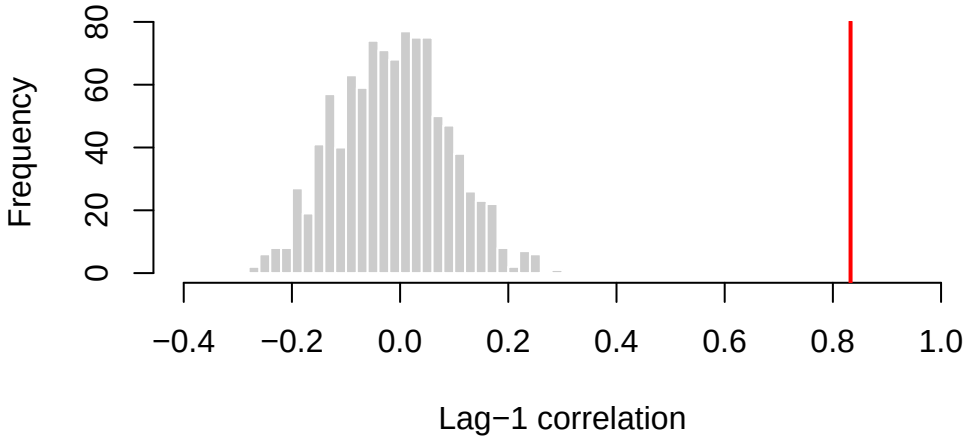


Figure 5: Lag-1 correlation for 1000 permutations of seis00 (histogram) and for the observed data (red line).

If the data were exchangeable, the observed order would be as likely as any permutation, and $\hat{\rho}_1$ would look like the permuted values. The observed lag-1 correlation is $\hat{\rho}_1 = 0.83$, while the permuted values lie in $[-0.28,\ 0.29]$. None of the 1000 permutations are as extreme ($p < 0.001$), so the claim of exchangeability is rejected. Neighbouring samples are strongly correlated, as expected from $\mathbf{d} = \mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}$.

Part 2

2a)

For a given (T, B) only the prior mean changes, to $\boldsymbol{\mu}_m^{T,B}$; the covariance is as in Part 1. From 1a, the marginal distribution of the data is

$$p(\mathbf{d} \mid T, B) = N\big(\mathbf{d};\ \mathbf{A}\boldsymbol{\mu}_m^{T,B},\ \boldsymbol{\Sigma}_d\big), \qquad \boldsymbol{\Sigma}_d = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I},$$

and Bayes' rule gives, for $(T, B) \in \mathcal{G}$,

$$p(T, B \mid \mathbf{d}) = \frac{p(\mathbf{d} \mid T, B)\, p(T, B)}{\sum_{(T', B') \in \mathcal{G}} p(\mathbf{d} \mid T', B')\, p(T', B')} \ \propto\ \exp\Big\{ -\tfrac{1}{2}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B})^\top \boldsymbol{\Sigma}_d^{-1} (\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m^{T,B}) \Big\},$$

since the prior $p(T, B) = 1/|\mathcal{G}|$ and the normalising constant of the Gaussian do not depend on (T, B). Here $|\mathcal{G}| = 751$ pairs.

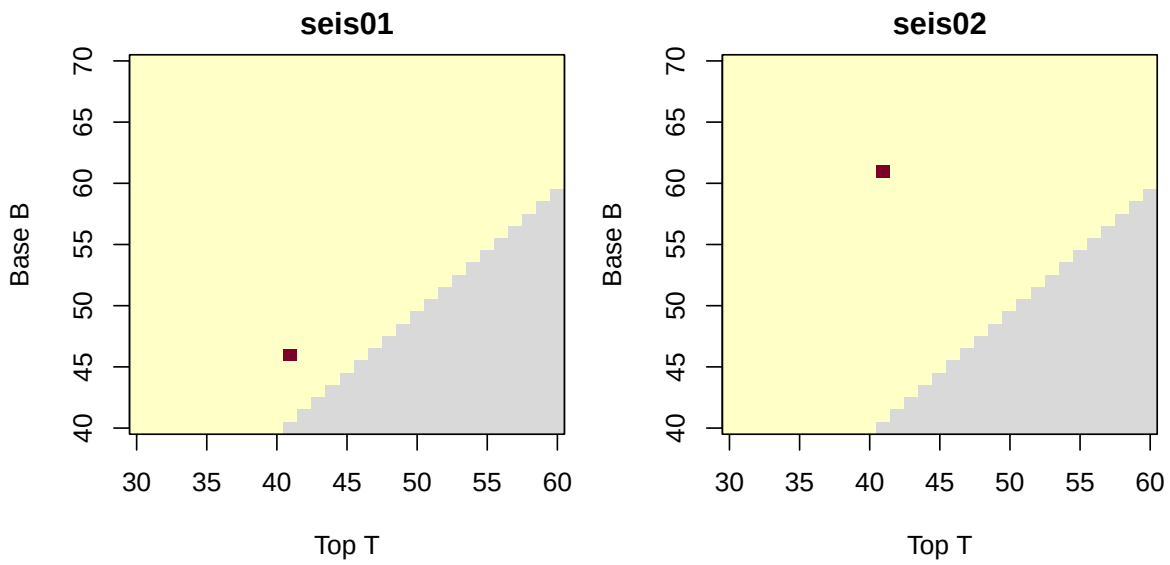


Figure 6: Joint posterior $p(T, B \mid \mathbf{d})$ for seis01 (left) and seis02 (right). Grey: inadmissible pairs.

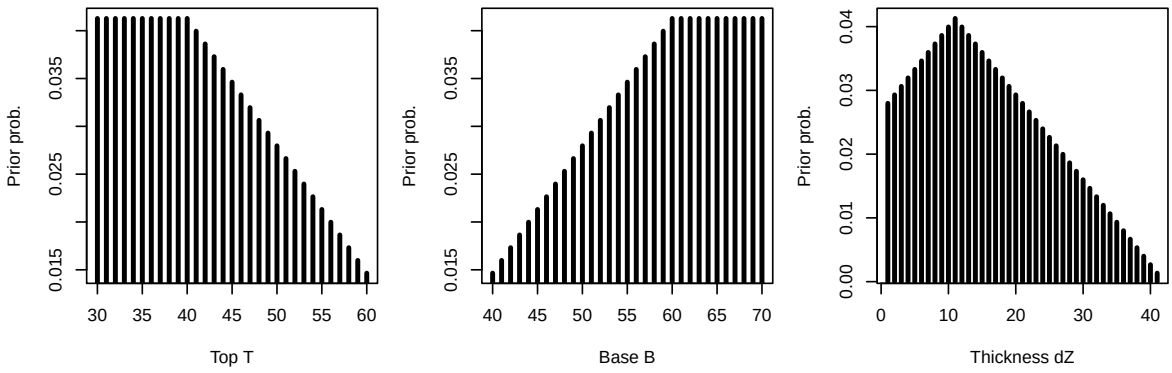


Figure 7: Marginal prior of top T, base B and thickness ΔZ (uniform prior on $\mathcal{G}$).

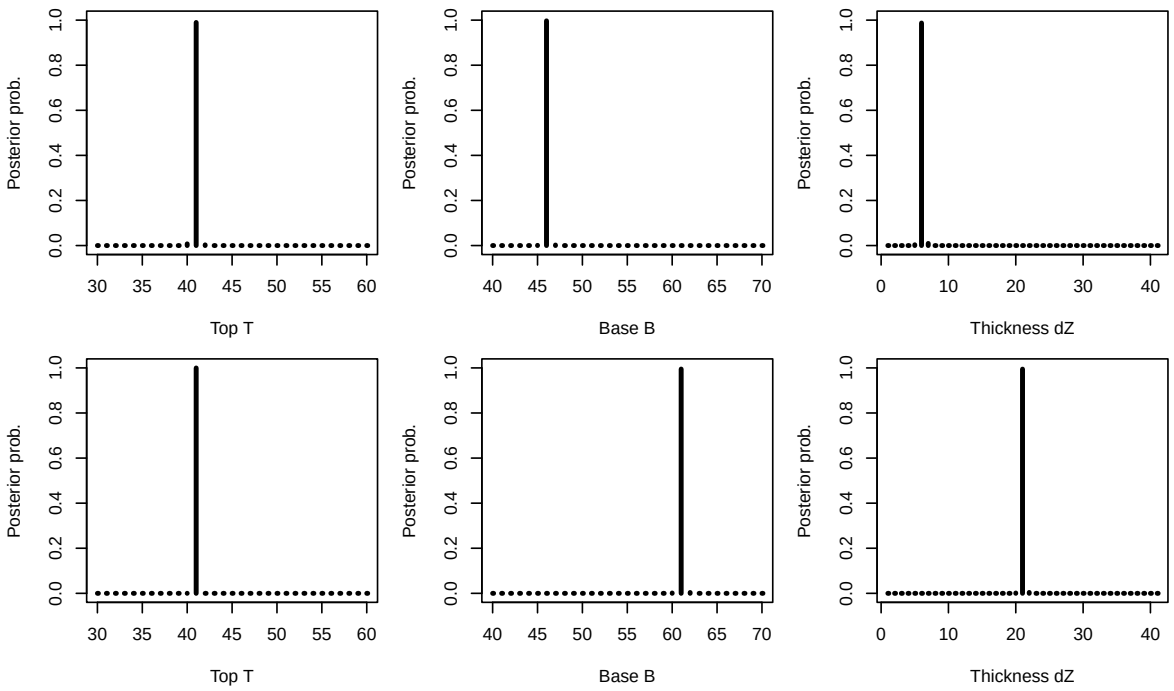


Figure 8: Marginal posterior of T, B and ΔZ. Top row: seis01, bottom row: seis02.

Table 2: Posterior mode of (T, B), its probability, and posterior mean thickness.

	T	B	dZ	posterior prob.	E(dZ | d)
seis01	41	46	6	0.987	6.01
seis02	41	61	21	0.995	21.00

The uniform prior on $\mathcal{G}$ does not give uniform marginals, because the number of admissible partners varies. T is uniform up to 40 and then decreasing, B is increasing up to 50 and then uniform, and ΔZ is most likely around 11 and least likely for very thick zones. The data overrule this completely. The posterior is concentrated on almost a single pair: a zone from 41 to 46 ($\Delta Z = 6$) for seis01, and from 41 to 61 ($\Delta Z = 21$) for seis02. The remaining uncertainty is at most one sample in T or B. A shift of $\Delta\mu = 5$ is large compared with $\sigma_m = 1$ and gives strong reflections at the top and base of the zone, so they are easy to locate.

2b)

The no-zone case is a single model ($\boldsymbol{\mu}_m = \mu\mathbf{1}$, as in Part 1), while the zone case consists of $|\mathcal{G}| = 751$ models. Giving every one of the 752 models the same prior probability would give $P(\text{no zone}) = 1/752$. That would say we are almost certain a zone exists, only because there are many ways to place it. The prior probability of the zero case should instead express our belief about whether a zone is present at all, and the remaining probability should then be spread over $\mathcal{G}$. We use

$$P(\text{no zone}) = \tfrac{1}{2}, \qquad p(T, B) = \tfrac{1}{2} \cdot \tfrac{1}{|\mathcal{G}|},\ (T, B) \in \mathcal{G}.$$

Then

$$P(\text{no zone} \mid \mathbf{d}) = \frac{\frac{1}{2}\, p(\mathbf{d} \mid \text{no zone})}{\frac{1}{2}\, p(\mathbf{d} \mid \text{no zone}) + \frac{1}{2}\, \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} p(\mathbf{d} \mid T, B)}.$$

The posterior probability of zero thickness is 2.1×10^{-8} for seis01 and 3.4×10^{-13} for seis02. The data are so much more likely under the zone models that both data sets clearly indicate a zone. Any reasonable choice of $P(\text{no zone})$ gives the same conclusion.

2c)

Let the action be a_1 (invest) or a_0 (do not invest), with loss equal to minus the profit:

$$L(a_1, V) = -(aV - b), \qquad L(a_0, V) = 0.$$

The Bayes decision minimises the posterior expected loss: $E\{L(a_1, V) \mid \mathbf{d}\} = -(aE(V \mid \mathbf{d}) - b)$ and $E\{L(a_0, V) \mid \mathbf{d}\} = 0$. Hence

$$\text{invest} \iff E(V \mid \mathbf{d}) > \frac{b}{a} = 10.$$

V is the exceedance volume from 1d. Its posterior combines the uncertainty in the zone and in $\mathbf{m}$: $p(V \mid \mathbf{d}) = \sum_{\text{models}} P(\text{model} \mid \mathbf{d})\, p(V \mid \mathbf{d}, \text{model})$. We simulate a model (no zone or (T, B)) from its posterior (2b), and then $\mathbf{m} \mid \mathbf{d}, T, B \sim N(\boldsymbol{\mu}_{m|d}^{T,B}, \boldsymbol{\Sigma}_{m|d})$ from 1a with prior mean $\boldsymbol{\mu}_m^{T,B}$, and compute $V = h(\mathbf{m})$.

Table 3: Posterior summaries of V and the profit $aV - b$ (5000 simulations).

	E(V|d)	E(profit|d)	P(profit<0|d)	P10	P50	P90
seis01	31.17	21.17	0	11.66	19.80	32.75
seis02	89.56	79.56	0	60.69	78.71	99.45

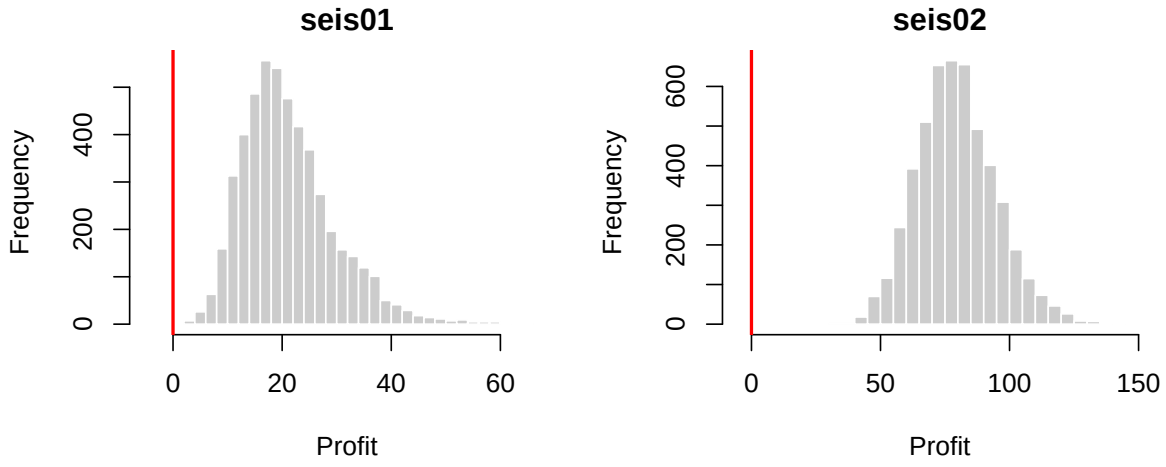


Figure 9: Posterior distribution of the profit $V - 10$. Red line: break-even.

Recommendation. For both data sets $E(V \mid \mathbf{d})$ is far above $b/a = 10$, so the recommendation is to invest in both cases. None of the 5000 simulations gives a loss in either case ($P(\text{profit} < 0 \mid \mathbf{d}) < 1/5000$). The expected profit is much larger for seis02 (thicker zone). To the decision makers I would present the uncertainty as the profit distribution (histograms above) with P_{10}, P_{50}, P_{90} and the probability of a loss, rather than a single number.

If V is instead read as the thickness ΔZ, the rule becomes $E(\Delta Z \mid \mathbf{d}) > 10$. This gives $E(\Delta Z \mid \mathbf{d}) = 6$ (do not invest) for seis01 and 21 (invest) for seis02.

Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
A  <- as.matrix(read.table("data/A.txt"))   # forward operator (n x n)
dimnames(A) <- NULL
d0 <- scan("data/seis00.txt", quiet = TRUE)
d1 <- scan("data/seis01.txt", quiet = TRUE)
d2 <- scan("data/seis02.txt", quiet = TRUE)
n  <- nrow(A)
z  <- 1:n
# Prior covariance: exponential correlation function
exp_cov <- function(n, sigma2, ell) {
  D <- abs(outer(1:n, 1:n, "-"))           # matrix of |i - j|
  sigma2 * exp(-3 * D / ell)
}

# Posterior of m | d for d = A m + eps (Gaussian conditioning)
gauss_posterior <- function(d, A, mu, Sigma, sigma2_eps) {
  Sigma_d <- A %*% Sigma %*% t(A) + sigma2_eps * diag(nrow(A))   # Cov(d)
  K       <- Sigma %*% t(A) %*% solve(Sigma_d)                   # Cov(m,d) Cov(d)^-1
  mu_post <- drop(mu + K %*% (d - A %*% mu))
  S_post  <- Sigma - K %*% A %*% Sigma
  S_post  <- (S_post + t(S_post)) / 2      # make exactly symmetric (rounding)
  list(mean = mu_post, cov = S_post)
}

# Simulate B draws from N(mu, S): m = mu + L x, L L^T = S, x ~ N(0, I)
rmvn <- function(B, mu, S) {
  L <- t(chol(S + 1e-10 * diag(length(mu))))   # lower Cholesky factor (tiny jitter)
  X <- matrix(rnorm(length(mu) * B), length(mu), B)
  t(mu + L %*% X)                               # one draw per row
}

# Mean with pointwise 95% credibility band (and optional realisations)
band_plot <- function(mean, sd, main, ylim, sims = NULL) {
  lo <- mean - 1.96 * sd
  hi <- mean + 1.96 * sd
  plot(z, mean, type = "n", ylim = ylim, xlab = "Index i", ylab = "m_i", main = main)
  polygon(c(z, rev(z)), c(lo, rev(hi)), col = "grey85", border = NA)
  if (!is.null(sims)) matlines(z, t(sims), lty = 1, lwd = 1,
                               col = c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e"))
  lines(z, mean, lwd = 2)
}
mu <- 0; sigma2_m <- 1; ell <- 20; sigma2_eps <- 0.04
mu_m    <- rep(mu, n)
Sigma_m <- exp_cov(n, sigma2_m, ell)

set.seed(1)
prior_sims <- rmvn(5, mu_m, Sigma_m)
band_plot(mu_m, sqrt(diag(Sigma_m)), "Prior", ylim = c(-3.5, 3.5), sims = prior_sims)
post0 <- gauss_posterior(d0, A, mu_m, Sigma_m, sigma2_eps)
```

```r
sd0   <- sqrt(diag(post0$cov))
band_plot(post0$mean, sd0, "Posterior given seis00", ylim = c(-3.5, 3.5))
abline(h = c(-1.96, 1.96), lty = 2, col = "grey40")
cov_image <- function(S, main, zlim) {
  image(z, z, t(S)[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "j", ylab = "i", main = main, axes = FALSE, useRaster = TRUE)
  axis(1); axis(2, at = n + 1 - c(1, 20, 40, 60, 80, 100), labels = c(1, 20, 40, 60, 80, 100))
  box()
}
zl <- range(Sigma_m, post0$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cov_image(Sigma_m, "Prior covariance", zl)
cov_image(post0$cov, "Posterior covariance", zl)
sd_range <- range(sd0)
A_one    <- max(abs(A %*% rep(1, n)))      # A applied to a constant vector
c_lev <- -1
h_V   <- function(m, c = c_lev) sum((c - m) * (m < c))   # exceedance volume

B <- 100
set.seed(2)
post_sims <- rmvn(B, post0$mean, post0$cov)
V <- apply(post_sims, 1, h_V)                              # one V per posterior draw

V_summary <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_summary, 2)), caption = "Posterior summaries of $V$ ($B=100$).")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
Vs  <- sort(V)                                          # order statistics
P10 <- Vs[10]
P90 <- Vs[90]

# N = number of samples below the true P10 ~ Bin(B, 0.1)
# V_(r) <= P10 < V_(s)  <=>  r <= N <= s - 1
r <- qbinom(0.025, B, 0.1)
s <- qbinom(0.975, B, 0.1) + 1
coverage <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)
lag1 <- function(x) cor(x[-1], x[-length(x)])   # lag-1 correlation

rho_obs <- lag1(d0)
set.seed(4)
rho_perm <- replicate(1000, lag1(sample(d0)))    # sample() gives a random permutation
n_extreme <- sum(abs(rho_perm) >= abs(rho_obs))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = c(-0.4, 1))
abline(v = rho_obs, col = "red", lwd = 2)
dmu <- 5
G   <- subset(expand.grid(T = 30:60, B = 40:70), T <= B)    # admissible (T, B)
G$dZ <- G$B - G$T + 1                                       # thickness

# Prior mean for a zone from T to B (T = NA: no zone)
mu_TB <- function(T, B) {
  m <- rep(mu, n)
  if (!is.na(T)) m[T:B] <- mu - dmu
```

```
  m
}

Sigma_d <- A %*% Sigma_m %*% t(A) + sigma2_eps * diag(n)
L_d     <- t(chol(Sigma_d))                                  # Sigma_d = L_d L_d^T

# log p(d | mean) up to a constant common to all models
loglik <- function(d, mean_m) {
  r <- forwardsolve(L_d, d - A %*% mean_m)                   # L_d^-1 (d - A mu)
  -0.5 * sum(r^2)
}

log_sum_exp <- function(x) max(x) + log(sum(exp(x - max(x))))

zone_posterior <- function(d) {
  ll <- sapply(seq_len(nrow(G)), function(k) loglik(d, mu_TB(G$T[k], G$B[k])))
  list(ll = ll, post = exp(ll - log_sum_exp(ll)))            # uniform prior cancels
}

zp1 <- zone_posterior(d1)
zp2 <- zone_posterior(d2)
heat <- function(post, main) {
  M <- matrix(NA, 31, 31, dimnames = list(30:60, 40:70))
  M[cbind(G$T - 29, G$B - 39)] <- post
  image(30:60, 40:70, M, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "Top T", ylab = "Base B", main = main)
  image(30:60, 40:70, ifelse(is.na(M), 1, NA), col = "grey85", add = TRUE)
  box()
}
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
heat(zp1$post, "seis01")
heat(zp2$post, "seis02")
marg <- function(w, v, levels) sapply(levels, function(x) sum(w[v == x]))
prior_G <- rep(1 / nrow(G), nrow(G))
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(30:60, marg(prior_G, G$T,  30:60), type = "h", lwd = 3, xlab = "Top T",  ylab = "Prior prob.")
plot(40:70, marg(prior_G, G$B,  40:70), type = "h", lwd = 3, xlab = "Base B", ylab = "Prior prob.")
plot(1:41,  marg(prior_G, G$dZ, 1:41),  type = "h", lwd = 3, xlab = "Thickness dZ", ylab = "Prior prob.
par(mfrow = c(2, 3), mar = c(4, 4, 1, 1))
for (zp in list(zp1, zp2)) {
  plot(30:60, marg(zp$post, G$T,  30:60), type = "h", lwd = 3, ylim = c(0, 1), xlab = "Top T",  ylab =
  plot(40:70, marg(zp$post, G$B,  40:70), type = "h", lwd = 3, ylim = c(0, 1), xlab = "Base B", ylab =
  plot(1:41,  marg(zp$post, G$dZ, 1:41),  type = "h", lwd = 3, ylim = c(0, 1), xlab = "Thickness dZ", yl
}
summ <- function(post) {
  k <- which.max(post)
  c(T = G$T[k], B = G$B[k], dZ = G$dZ[k], prob = post[k], E_dZ = sum(post * G$dZ))
}
tab2a <- rbind(seis01 = summ(zp1$post), seis02 = summ(zp2$post))
knitr::kable(tab2a, digits = c(0, 0, 0, 3, 2),
             col.names = c("T", "B", "dZ", "posterior prob.", "E(dZ | d)"),
             caption = "Posterior mode of $(T,B)$, its probability, and posterior mean thickness.")
p0 <- 1 / 2
post_zero <- function(zp, d) {
```

```
  l0 <- log(p0)     + loglik(d, mu_TB(NA, NA))
  lz <- log(1 - p0) + log_sum_exp(zp$ll) - log(nrow(G))
  exp(l0 - log_sum_exp(c(l0, lz)))
}
pz <- c(seis01 = post_zero(zp1, d1), seis02 = post_zero(zp2, d2))
a <- 1; b <- 10; B_mc <- 5000
K_gain <- Sigma_m %*% t(A) %*% solve(Sigma_d)
L_post <- t(chol(post0$cov + 1e-10 * diag(n)))             # same posterior covariance as 1c

sim_V <- function(zp, d, pz0) {
  w   <- c(pz0, (1 - pz0) * zp$post)                       # no zone + all (T, B)
  Tm  <- c(NA, G$T); Bm <- c(NA, G$B)
  idx <- sample(length(w), B_mc, replace = TRUE, prob = w)
  sapply(idx, function(k) {
    mu_k <- mu_TB(Tm[k], Bm[k])
    m    <- mu_k + K_gain %*% (d - A %*% mu_k) + L_post %*% rnorm(n)
    h_V(m)
  })
}
set.seed(5)
V1 <- sim_V(zp1, d1, pz[1])
V2 <- sim_V(zp2, d2, pz[2])
profit <- list(seis01 = a * V1 - b, seis02 = a * V2 - b)

tab2c <- t(sapply(profit, function(x)
  c(EV = mean(x + b) / a, E_profit = mean(x), P_loss = mean(x < 0),
    P10 = unname(quantile(x, 0.1)), P50 = median(x), P90 = unname(quantile(x, 0.9)))))
knitr::kable(tab2c, digits = 2,
             col.names = c("E(V|d)", "E(profit|d)", "P(profit<0|d)", "P10", "P50", "P90"),
             caption = "Posterior summaries of $V$ and the profit $aV-b$ (5000 simulations).")
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
for (nm in names(profit)) {
  hist(profit[[nm]], breaks = 30, col = "grey80", border = "white", main = nm,
       xlab = "Profit", xlim = range(-5, profit[[nm]]))
  abline(v = 0, col = "red", lwd = 2)
}
```